SUMMARY WEEK 4

JSON vs CSV Concept Map

A Comprehensive Technical Comparison for Data Structuring & Representation

JSON

JavaScript Object Notation

STRUCTURE & FORMAT

- Hierarchical and multi-level nested data.
- Organized into Key-Value pairs & Arrays.
- Flexible schema (supports complex structures).

DATA EXAMPLE

```
{
  "user": "Alice",
  "age": 25,
  "skills": ["Python", "SQL"]
}
```

SUPPORTED TYPES

- Strings, Numbers, Booleans, Null.
- Complex types: Arrays `[]` and Objects `{}`.

PRIMARY USE CASES

- Web & Mobile APIs / RESTful Services.
- NoSQL Database Documents (e.g., MongoDB).
- Configuration & metadata files.

CSV

Comma-Separated Values

STRUCTURE & FORMAT

- Flat 2D tabular layout (Rows & Columns).
- Delimited by commas or specific separators.
- Fixed schema (every record shares header fields).

DATA EXAMPLE

```
Name,Age,Skill
Alice,25,Python
Bob,30,SQL
```

SUPPORTED TYPES

- Plain text and numeric representations.
- No native array or object support.

PRIMARY USE CASES

- Data analysis & Spreadsheets (Excel / Pandas).
- Bulk database exports & data ingestion.
- Simple tabular dataset transfers.

Key Differences Matrix

Feature	JSON	CSV
Data Model	Hierarchical / Tree / Nested	Flat / Two-Dimensional Table
	Higher (Keys repeated per object)	

Feature	JSON	CSV
File Size & Overhead		Lower (Headers declared once at top)
Readability	Human-readable with proper formatting	Extremely clean in spreadsheet tools
Flexibility	High (Dynamic structures & missing fields)	Low (Strict row and column structure)